

SHARED ENVIRONMENT

SUMO traffic environment

TraCI state extraction

Vehicle-level actions

PHASE 1 - COMPLETE PIPELINE EVALUATION

Rule-Based | Raw LLM | Hybrid | Hybrid + Safety

Structured traffic state

LLM provider and parser

Cooperative processing

Deterministic safety authority

PROCEED / WAIT / FREE

Fallback

PHASE 2 - MATCHED SAFE-CANDIDATE COMPARISON

Privacy-minimised local state

Route semantics and compatibility

Same safe candidate groups

Identical planner action space

Deterministic cooperative comparator

Gemini candidate selector

Passage grant lifecycle

Final deterministic safety

Decision provenance and attribution evidence
provider | parser | fallback | candidates | planner | safety | executed action | outcome